

UNIVERSITY OF SOUTHAMPTON
Faculty of Engineering and Physical Sciences
School of Electronics and Computer Science

A progress report submitted for continuation towards a MEng

Supervisor: Dr BooJoong Kang
Examiner: ???

**A Reproducible Evaluation of Linux
Rootkit Detectors via the Implementation
of an Automated Testbed &
Proof-of-Concept Rootkit**

by **Elliott Jelbert**

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ABSTRACT

FACULTY OF ENGINEERING AND PHYSICAL SCIENCES
SCHOOL OF ELECTRONICS AND COMPUTER SCIENCE

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This work is all about ...

Contents

Nomenclature	xiii
Acknowledgements	xv
1 Background	1
1.1 Linux & The Rootkit	1
1.2 Evolution of Rootkit Techniques	1
1.3 Rootkit Detectors	1
2 Literature Review	3
2.1 Hi	3
Bibliography	5

List of Figures

List of Tables

Listings

Nomenclature

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Acknowledgements

Thanks to no one.

To ...

Chapter 1

Background

1.1 Linux & The Rootkit

Rootkits are typically described as a specialised piece of malware built for persistence & stealth which is installed *after* an attacker has gained root-level access to a system [1]. They may also give an attacker additional functionality such as network backdoors (cite magic bpf), the ability to hide files or other capabilities that give the attacker a greater ability to hide their presence whilst maintaining persistence.

Linux & subsequently Linux rootkits have been chosen for this project due to its relevance and prevalence in cloud & server environments. Additionally, a wide range of rootkit techniques exist for Linux which will be covered in the following section. Consequently, many tools have been developed which aim to detect the presence of a rootkit on a system. To achieve this, these tools may take a variety of approaches which will be covered in section 1.3.

The reliability and accuracy of these tools against current rootkits is paramount to ensure that test test...

1.2 Evolution of Rootkit Techniques

Linux rootkits have had an evolution

1.3 Rootkit Detectors

Chapter 2

Literature Review

2.1 Hi

hello

Bibliography

- [1] National Institute of Standards and Technology, “Rootkit,” Computer Security Resource Center Glossary, n.d., accessed: 07-10-2026. [Online]. Available: <https://csrc.nist.gov/glossary/term/rootkit>